

GH - 300

Pull Requests & Code Review with Copilot

PR summaries, AI-assisted review, and the Cloud Coding Agent that closes the loop.

PR Summary

AI Code Review

Custom Instructions

Cloud Agent

Protected branches

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Deck 5 · GH-300 Certified GitHub Copilot Expert Training

Three Copilot capabilities in the PR workflow

PR Summary

Auto-generated description + title from the diff. Lets reviewers find what changed and why before they open the first file.

AI Code Review

Copilot reviews the PR with full project context. Suggests fixes you can apply with one click. Same engine in IDE and on github.com.

Cloud Coding Agent

Assign an issue to Copilot — it researches, codes, runs CI, self-reviews, and opens a draft PR. You drive the merge.

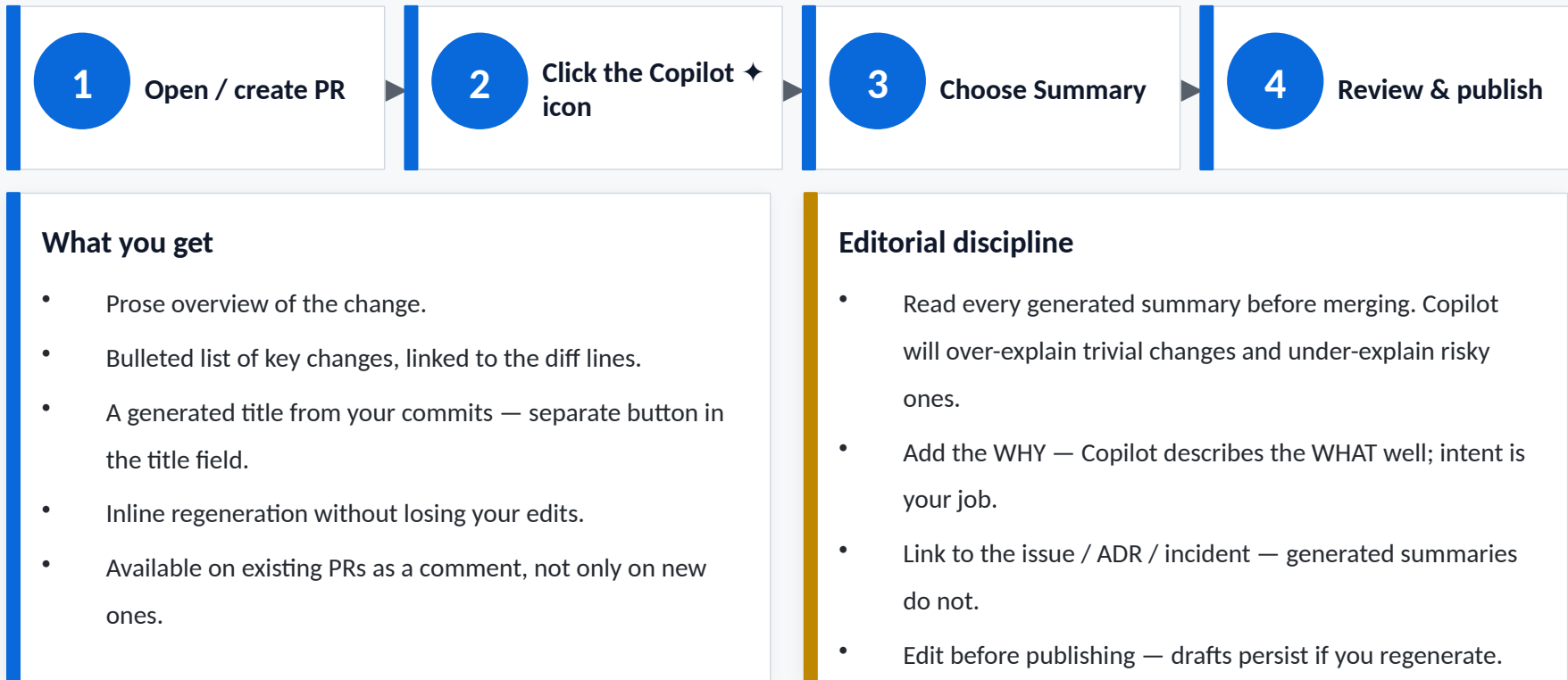
Honest framing

Supplement, not substitute. Copilot Code Review is a second-pair-of-eyes pass; it does not replace human judgment on intent, business correctness, or architecture.

It misses things. Context is bounded by the model window and time budget — large PRs get partial coverage. Pair with human review for production paths.

Speed without quality is debt. Use Copilot to find more — not to merge more.

PR Summary — auto-generated descriptions and titles



AI Code Review — how it works

Full project context

Reads the diff plus related files, cross-file dependencies, the PR description, and commit messages.

Multi-angle analysis

Correctness, security, readability, performance — in any language Copilot supports. Same engine in IDE and github.com.

Suggested fixes

One-click commits for accepted suggestions. Batch multiple suggestions into a single commit.

Auto-fix via Cloud

Hand a review batch to the Cloud Coding Agent. It opens a new PR with the fixes — closing the loop.

Code Review — step by step on github.com

1

Open your PR

Create or navigate to an existing pull request.

2

Request Copilot review

Reviewers menu → select Copilot.

3

Wait

Usually < 30 seconds for a small PR.

4

Read comments

Copilot leaves a "Comment" review — never Approve / Request Changes.

5

Apply suggestions

Single fix → commit, or batch multiple → one commit.

6

Iterate

React, reply, or resolve — just like a human review.

Note: Copilot reviews do not count toward required approvals by default — but rulesets can be configured to weigh them in for non-protected branches. Verify your specific branch policy before relying on this behaviour.

Code Review in your IDE — shift left, save the round-trip

VS Code

- Select code → right-click → Generate Code → Review.
- Or ask in Copilot Chat: "review this code" with #selection.
- Comments appear inline + in the Comments panel.
- Honors .github/copilot-instructions.md.

Visual Studio

- Requires Visual Studio 17.14 or later.
- Select code in the editor → request review.
- Same review quality as github.com.
- Custom instructions sourced from base branch.

NEW • Review whole PRs from the VS Code Pull Requests panel

- Open the GitHub Pull Requests extension — review the full PR diff without leaving the editor.
- Request a Copilot review from inside the panel — comments stream back into VS Code.
- @copilot in PR comments triggers iteration; commits flow back to the branch automatically.
- Custom-instruction overrides apply per-branch — useful when the base branch has different style rules.

Automatic Code Review — configure once, review everything

For yourself

Copilot Settings → Automatic Copilot code review → Enabled. Requires a paid Copilot plan.

For a repository

Repo Settings → Rules → Rulesets → branch ruleset → "Automatically request Copilot code review" — now a standalone rule with Run on each push and Run on drafts sub-settings.

For the org

Org Settings → Copilot → Features → enable Automatic Copilot code review. Set review effort level and a default Actions runner org-wide.

⚠ Billing — live since June 1, 2026

What changed: All Copilot plans now bill on GitHub AI Credits, and each code review also consumes GitHub Actions minutes on private repos. Public repos stay free.

Why this matters: Auto-review-on-every-push across many repos can spike Actions minute usage. Audit your rulesets and set a budget cap.

Admin lever: Org admins can now set a default Actions runner for code review across all repos — no per-repo setup. That is the practical answer to the cost concern.

Recommendation: Enable review on draft PRs; consider disabling review-on-every-push if your team rebases frequently.

Custom instructions for code review — the highest-leverage quality knob

How it works

The repo or org "Code review instructions" box, plus copilot-instructions.md, *.instructions.md, and AGENTS.md. The 4,000-character cap was removed in June 2026.

What to put there

Only what differs from your code style guide. Be specific about what the reviewer should focus on — and what it should NOT comment on (formatting handled by Prettier / dotnet-format, etc.).

review section

```
# Code review focus for this repo
```

```
Project: payments service · runtime: .NET 8
```

Focus on:

- Concurrency: new async paths must propagate CancellationToken end-to-end.
- Money: decimal only — flag any double / float arithmetic on monetary values.
- Audit: every state transition on Order must emit an AuditEvent before return.
- PII: do not log raw card numbers; mask to last 4 digits using Mask.LastFour().

Do NOT comment on:

- Formatting — handled by dotnet-format.
- Naming on private members — _camelCase is intentional.

Cloud Coding Agent — from issue to pull request

1

Assign issue

Assignees → Copilot.
Optional prompt for guidance.

2

Research + plan

Reads the repo, related files, and the issue body.
Drafts an internal plan.

3

Code + test

Edits in an ephemeral env (GitHub Actions).
Runs tests + linters, auto-corrects.

4

Draft PR

Opens a PR and requests your review. @copilot in comments to iterate.

5

Merge

You merge. Or push your own commits on the same branch.

Triggers from anywhere — github.com is not the only entry point

- github.com — Assignees → Copilot on any issue.
- GitHub CLI — `gh agent-task create` / `gh copilot agent` for non-interactive dispatch (verify the verb against current docs).
- VS Code · Visual Studio · GitHub Mobile · MCP-enabled IDEs — same agent, surfaced where you work.

Reviewing a Copilot-created PR — five moves

@copilot in comments

Mention @copilot to ask for changes. It adds a 🤖 reaction and pushes commits to the branch.

Batch your feedback

Submit review comments together; Copilot responds to comments from anyone with write access.

Choose a model

Use the model picker on github.com when submitting review comments. Default = the model that created the PR. (Updated Apr 2026.)

Push your own changes

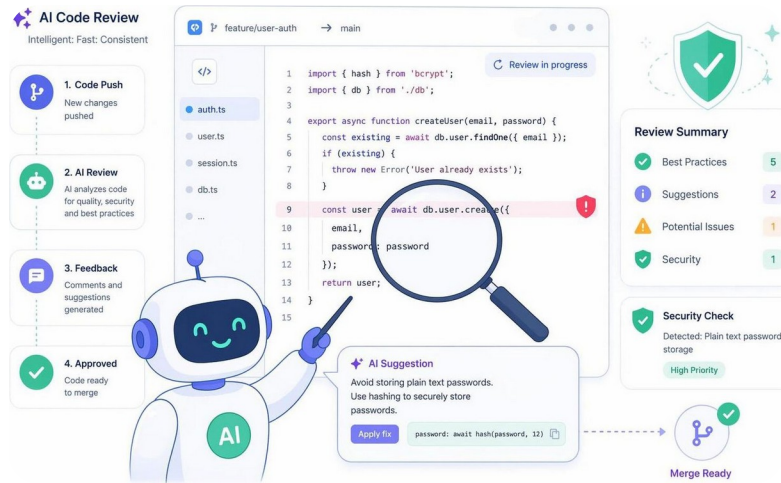
Check out the branch and edit yourself. Mix human and AI work on one PR.

Request a separate PR

Ask Copilot in plain language to open a new PR for fixes instead of pushing here.

Cloud Agent self-review + security scan — closing the loop

- 1 Draft**
Agent writes code and tests in the ephemeral env.
- 2 Self-critique**
Re-reads the diff as a reviewer would. Looks for the issues a human reviewer would raise first.
- 3 Security scan**
Runs CodeQL / dependency review on the diff. Flags injection, hardcoded secrets, vulnerable packages.
- 4 Self-fix**
Applies the resulting suggestions automatically — same engine as Copilot Code Review.
- 5 Draft PR**
PR opened with a self-summary and the list of issues already fixed. Human review starts higher up the stack.



What this changes for human reviewers

The first round of trivial findings (lint, obvious null-derefs, common security smells) is already cleared before the PR opens. Your job moves up the stack: intent, business correctness, architectural fit. Read the agent's self-summary — it tells you which issues were already addressed.

What Copilot checks — and what it cannot

Correctness

Logic errors, off-by-one, null refs, unhandled edges.

Security

Injection, unsafe data handling, auth gaps, hardcoded secrets.

Readability

Naming, complexity, dead code, magic numbers.

Performance

Inefficient loops, N+1 queries, missing memoisation.

Style & standards

Project conventions from copilot-instructions.md and AGENTS.md.

Cross-file impact

How the diff affects callers; breaking changes flagged.

Limitations to remember

- Not guaranteed to catch all issues — always pair with human review.
- Large PRs get partial coverage — bounded by model window and time budget.
- Cannot replace domain expertise — business correctness is still yours.
- Will not catch issues outside the diff unless explicitly referenced.

Protected branches + required reviewers — how Copilot fits

Default behaviour

- Copilot reviews are "Comment" reviews — never Approve / Request Changes.
- They never block merging on protected branches.
- They do not count toward required approvals.
- Your own approval on a Copilot-created PR does not count if you requested the agent.

Rulesets can do more

- Require Copilot review on every push to a ruleset-matched branch.
- Run review on draft PRs to catch errors before human review starts.
- Some experimental rulesets allow weighing Copilot review as N-of-required for non-prod branches — verify before relying on it.

Practical policy for most teams

Production branches (main, release/*): 1+ human approval required, Copilot review enabled but advisory.

Feature branches: Copilot review on draft PRs (cheap signal, catches issues before human time).

Bot PRs (dependabot, renovate): Auto-review on every push; merge only after green CI + Copilot has no error-severity comments.

Cloud-Agent-created PRs: Always require at least one human approval — self-review is supplement, not replacement.

Key takeaways

1

Three capabilities, one workflow

PR Summary, AI Code Review, Cloud Coding Agent — same project context, same custom instructions.

2

Specific review instructions win

The 4K-char review prompt is the highest-leverage knob for review quality. Tell Copilot what to focus on AND what to ignore.

3

Shift review left

Review in the IDE (and the VS Code PR panel) before opening the PR — cheaper signal, faster loop.

4

Cloud agent closes the loop

Self-review + security scan before the PR opens. Human review starts higher up the stack.

5

Mind the June 2026 billing change

Auto-review-on-every-push moves to GitHub Actions minutes. Audit rulesets and set budgets before cutover.

6

Copilot supplements humans

It will not catch everything. Always pair with a human reviewer on production paths.

Thank you.

Bring your messiest PR — we will review it together.

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